

Minimization of Maximum Mean Discrepancy (MMD)

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Annual student conference, UCL Computer Science, 2026

About Me



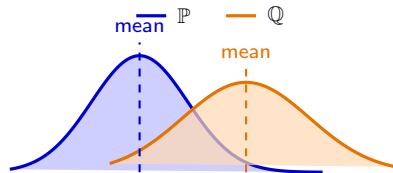
- Zonghao Chen
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- Graduated from Tsinghua University in 2022
 - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

Content: Minimization of MMD

- What is MMD?
- Minimization of MMD for numerical integration
- Minimization of MMD for sampling / generative modelling
- Minimization of MMD for statistical inference

Background: what is MMD?

- Given two distributions $\mathbb{P}$ and $\mathbb{Q}$, how to compare them?
- Compute the difference of their means: $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$.
- Compute the difference of their second moments: $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$.
- The first two moments are not enough!



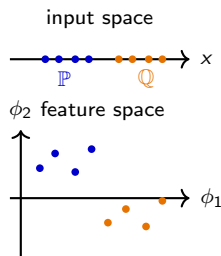
Goal: Compare $\mathbb{P}$ and $\mathbb{Q}$ with a general f , i.e., $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$?

Background: Maximum Mean Discrepancy (MMD)

- Compare $\mathbb{P}$ and $\mathbb{Q}$ by maximizing the difference over a function class $\mathcal{H}$:

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .
- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem 1

MMD admits a closed-form expression:

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

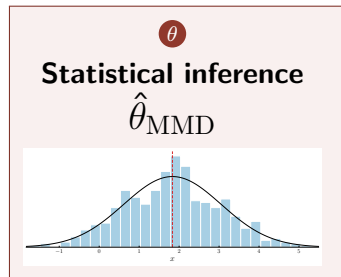
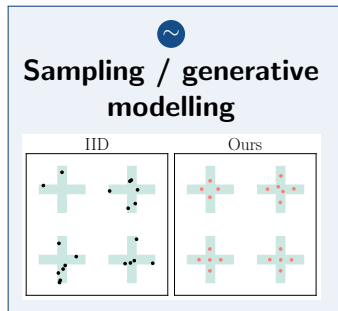
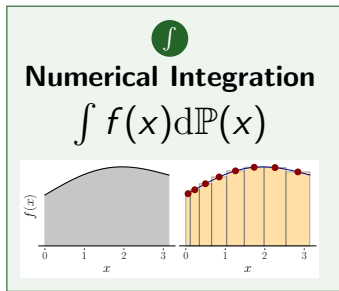
- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian).

Background: Maximum Mean Discrepancy (MMD)

We now have MMD

A kernel-based way to compare probability distributions $\mathbb{P}$ and $\mathbb{Q}$

What can we use it for?



Numerical Integration

Core task

Estimate expectations

$$\mathbb{E}_{\mathbb{P}}[f(X)] = \int f(x) d\mathbb{P}(x)$$

when analytic integration is unavailable.

- **Machine learning:** estimate risks, losses, and kernel expectations.
- **Bayesian inference:** compute posterior means and model evidence.
- **Reinforcement learning:** evaluate expected returns under stochastic policies.
- **Computer graphics:** render images via light transport integrals.
- **Probabilistic algorithms:** approximate sums, volumes, and randomized estimators.
- **Uncertainty quantification:** propagate randomness through simulators.

Numerical Integration

- Given a function f and a distribution $\mathbb{P}$, how to compute $\int f(x)d\mathbb{P}(x)$?
- Given a set of points $\{x_i\}_{i=1}^n$ and function evaluations $\{f(x_i)\}_{i=1}^n$.
- Quadrature rule:

$$\hat{I} = \sum_{i=1}^n w_i f(x_i).$$

- When $w_1 = \dots = w_n = 1/n$, this is Monte Carlo.
- $|\hat{I}_{MC} - I| = \mathcal{O}_P(N^{-1/2})$. Or equivalently, to obtain an ϵ -accurate estimate, we need $N = \mathcal{O}(\epsilon^{-2})$ samples.

Question: Can we do better than $\mathcal{O}(N^{-1/2})$?

